

STALLINGS FOLDINGS FOR RATIONAL SUBSETS OF AUTOMATIC GROUPS

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ABSTRACT. Let G be an automatic group with associated regular language L . We describe a procedure for constructing an automaton which recognises elements of a given submonoid or rational subset K of G . This builds on work of Kharlampovich, Miasnikov and Weil, on the case where K is a subgroup of G .

Our construction succeeds, after sufficiently many iterations, whenever K satisfies a certain convexity property, which we call L -proximity. We show how to test whether the construction is complete in the case that K is a submonoid; we have no such test for the general case of a rational subset K .

We focus particularly on the case of a surface group G of genus $g > 1$, where L is the language of geodesic words in the standard generators. We use small cancellation theory to obtain a method for constructing L -recognisable submonoids of G .

1. INTRODUCTION

This paper describes a procedure whose aim is to construct an automaton to recognise the elements of a specified submonoid or rational subset of an automatic group. The techniques involved are similar to those of the classic Stallings' folding algorithm, which applies to subgroups of free groups, and its recent generalisations. More precisely we develop the partial algorithm of [12, Section 4], which applies to a subgroup H of an automatic group G , with associated regular language L , and terminates with an automaton recognising H precisely when H is L -quasi-convex. Note that, since the procedure of [12] always terminates and produces the required output when the answer to the associated decision problem is 'yes' (but might fail to terminate otherwise), this procedure satisfies the criteria to be a partial algorithm.

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The concept of L -quasi-convexity, which is defined in [12], generalises the concept of quasi-convexity. We state the definition in Section 2.5 below, but note here that if G is hyperbolic and L is the set of all geodesics, then L -quasi-convexity is simply quasi-convexity.

Given an automatic group G , finitely generated by a set S , and with associated regular language L of words over S mapping onto G , [12] describes a partial algorithm to solve the membership problem for a finitely generated subgroup of G . If the finitely generated group is L -quasi-convex the partial algorithm always terminates, so is an algorithm. In this paper, we define a procedure that extends the partial algorithm of [12] to the setting of a rational subset K of G that may not be a subgroup. In order to terminate, our generalisation requires a condition on K that is stronger than L -quasi-convexity, which we call L -proximity (defined in Section 3). For an L -proximate rational subset K of G we construct a procedure which will, after a finite number of steps, output an automaton accepting the language of all words in L that map to K . In the case where K is an L -proximate submonoid of an automatic group G (necessarily also rational), we are able to turn this procedure into an algorithm, since we can test, at each iteration of the procedure, to see whether or not it has completed.

In Section 2 we make preliminary definitions and give some background on automata, rational subsets of groups, and convexity. In Section 3 we introduce the notion of an L -proximate subset of a group and establish the basic properties needed for our purposes. In addition, we describe a weaker L -proximity property that is sufficient to allow construction of a folding for finitely generated submonoids (with this property) of automatic groups. In Section 4 we describe our procedure to construct an automaton recognising the L -representatives of a rational subset of a group G . In Section 5 we discuss how and when it is possible to determine when our procedure has completed; for instance showing this for (weakly) L -proximate submonoids of automatic groups. In Section 6 we apply our results, and small cancellation theory, to submonoids of surface groups, and find some examples of submonoids with decidable membership problem which we believe are not covered in the literature.

In future work we address applications of our procedure to arbitrary rational subsets of certain groups (for example RAAGs).

1.1. Background: Stallings folding. The concepts of Stallings foldings and Stallings automata, when introduced in 1983 [15], provided group theorists with a completely new approach to understanding finitely generated subgroups of free groups.

Given a free group G with generating set S , and a subgroup $H = \langle T \rangle$ of G , specified by its finite generating set T of words over S , Stallings' algorithm applies a sequence of 'foldings' to a labelled graph constructed from T , iterating until it yields a deterministic finite state automaton (or labelled graph) $\text{St}(H)$, called the *Stallings automaton* of H , which recognises the set of all reduced words over S that belong to the subgroup H . From an automata theoretic point of view, the folding operations are precisely what is needed for determinisation and minimisation of a finite state automaton that has the property that the reverse of a transition on a symbol x is a transition on the symbol x^{-1} ; following [1] we shall call such an automaton *involutive*. From a graph theoretic point of view, $\text{St}(H)$ is the core of the Schreier graph of H .

The Stallings automaton for a subgroup provides more than just a finite object to recognise its elements; a lot of valuable information can be read from it. Information about the subgroup index, conjugates, normaliser, decidability problems regarding membership and intersections, separability and much more can be easily extracted from it.

The overall framework of a generalisation of Stallings folding to subgroups of groups beyond free groups entails a group G with finite generating set S and a finite set T of words over S , generating a subgroup $H \leq G$. With input a structure built from T , an operation (folding) is iteratively performed until a finite object $\text{St}(H)$, uniquely associated to H , and from which information about H can be extracted, is reached. The subgroup H needs to satisfy certain hypotheses to guarantee that this finite object is reached. For example, in the case of free groups H must be finitely generated. The work of Kharlampovich, Miasnikov and Weil [12], follows this framework to give a generalisation of Stallings foldings for subgroups of automatic groups, that have the L -quasi-convex property, thereby solving such problems as subgroup membership for this class of subgroups.

More recently, Dani and Levicovitz [5] used similar techniques to study subgroups of right-angled Coxeter groups, using a generalisation of folding that yields a Stallings-like object which is a cube complex instead of a graph. Information about normality and index is reflected in this cube complex and, in the case that the subgroup is finitely generated and quasi-convex, the associated cube complex is finite and this information can be explicitly read from that finite object. Generalising the results of [5], Ben-Zvi–Kropholler–Lyman [3] developed a Stallings foldings framework to study subgroups of fundamental groups of non-positively curved cube complexes. This construction also yields a finite object when the subgroup is finitely generated and quasi-convex, and this object too encodes information about normality and index, and allows solution of the subgroup membership problem.

2. DEFINITIONS AND NOTATION

2.1. Words, free monoids, free groups: Given a set Σ we denote by Σ^* the set of all words on Σ , as well as the free monoid over Σ if we want to stress the concatenation operation. A set Σ equipped with a map ν to a monoid M is called a *monoid generating set* for M if the natural extension of ν to a map from Σ^* to M is surjective. Usually all mention of ν is suppressed and often we implicitly assume that Σ is a subset of M and ν is the inclusion map. The definition of a *group generating set* for a group G is made the same way, using the free group on Σ instead of the free monoid on Σ . The symbols ε and 1 are respectively used to refer to the empty word and to the identity element (in a monoid or a group).

Throughout this document, unless stated otherwise, $S = \{s_1, \dots, s_r\}$ will denote a finite set (also referred to as an *alphabet* with elements called *letters*). The *formal inverse* of a word $w = s_1 \cdots s_n$, where $s_i \in S^\pm$, is the word $w^{-1} = s_n^{-1} \cdots s_1^{-1}$. For a subset T of $(S^\pm)^*$ we define the *formal inverse* of T to be $T^{-1} = \{w^{-1} : w \in T\}$. We denote by $S^\pm = S \sqcup S^{-1}$ the alphabet given by the letters in S together with their (formal) inverses. and by $(S^\pm)^*$ the set of all words over $S^\pm$. The free group on S , denoted $F(S)$, can be described on the set of equivalence classes of $(S^\pm)^*$ under the relation induced by free reduction, with concatenation as the group operation. In this way, $F(S)$ can be identified with the set of freely reduced words in $(S^\pm)^*$, and hence may be regarded as a subset of the free monoid over $S^\pm$. However

we note that this is an embedding of $F(S)$ in $(S^\pm)^*$ only as a set, and not as a submonoid.

The freely reduced representative of a word w in $(S^\pm)^*$ is denoted by $|\bar{w}|$. For a word w in $(S^\pm)^*$, we denote by $|w|$ its length, given by the number of letters in w , and by $|w|_{F(S)}$ the length $|\bar{w}|$ of $\bar{w}$.

For a group G with generating set S , we denote by $\mu: (S^\pm)^* \rightarrow G$ and $\pi: F(S) \rightarrow G$ the natural projections from the free monoid and the free group, respectively, onto G . Note that μ factors through π .

$$\begin{array}{ccc} (S^\pm)^* & \xrightarrow{\overline{(-)}} & F(S) \\ & \searrow \mu & \downarrow \pi \\ & & G = \langle S | R \rangle \end{array}$$

Understanding $F(S)$ as a subset of $(S^\pm)^*$ we can think of π as the restriction $\mu|_{F(S)}$ of μ to $F(S)$, and therefore of $\pi^{-1}(G)$ as a subset of $(S^\pm)^*$. To signify that two words w_1, w_2 in $(S^\pm)^*$ or $F(S)$ have the same image under μ or π , we will write $w_1 =_G w_2$.

Given a set of words $T \subset (S^\pm)^*$, we denote by $T^\pm$ the union of T with the set T^{-1} of its formal inverses; each element of the subgroup $\langle T \rangle$ can be represented as a product of elements of $T^\pm$. For w in $(T^\pm)^*$ (representing an element of the subgroup $\langle T \rangle$) we define the length of w in G with respect to T to be

$$|w|_{G,T} = \min\{n: w =_G v, v = h_1 \dots h_n, h_i \in T^\pm\}.$$

2.2. Finite state automata: We refer to [10] for notation and standard results, noting that more detail can be found in the classic text [11]. The notation $\mathcal{A} = (\Sigma, V, E, V_F, v_0)$ will be used to describe a *finite state automaton* (FSA) where Σ is a finite set (the *alphabet*), V is the set of vertices or *states*, E is the set of Σ -labelled edges or *transitions* of the form $(v, s, w) \in V \times \Sigma \times V$, $v_0 \in V$ is the *initial state* and $V_F \subset V$ is the set of *accepting states* of the automaton. As our notation suggests, we shall often view $\mathcal{A}$ as an edge-labelled directed graph. We say that (v, s, w) is an *s-labelled transition*, with v as its *source* state and w as its *target* state. For an edge $e = (v, s, w)$, we will use the notation $l(e) = s$ for the labelling function. Unless stated otherwise, the word automaton will implicitly refer to one with finitely many states, throughout this text.

To each finite state automaton $\mathcal{A}$ we can assign the set $L(\mathcal{A}) \subset \Sigma^*$ of words *accepted* by the automaton, which are the words given by the concatenation of edge labels of paths in $\mathcal{A}$ that start in v_0 and end in one of the accepting states; we call $L(\mathcal{A})$ the *language* accepted by $\mathcal{A}$, and say that $\mathcal{A}$ *recognises* $L(\mathcal{A})$. We say that a subset L of Σ^* is *regular* if it is the language $L(\mathcal{A}_L)$ accepted by a finite state automaton $\mathcal{A}_L$.

We can consider automata with ε -transitions, that is, automata over the alphabet $\Sigma_\varepsilon = \Sigma \sqcup \{\varepsilon\}$. We note that, given any automaton $\mathcal{A}_\varepsilon$ over the alphabet Σ_ε , an algorithm described in [10, Section 2.5.1] constructs an automaton $\mathcal{A}$ over Σ accepting the same language.

Similarly, given a *non-deterministic FSA* (one where s -labelled transitions (v, s, w) , (v, s, w') with different target states $w \neq w'$ are allowed) an algorithm described in [10, Proposition 2.5.2] constructs a *deterministic FSA*

From the above it is clear that there are many different FSA accepting a given regular set. However, there is a unique one that is deterministic and has the least number of states, which we call the *minimal automaton*, and there is an algorithm to obtain it from a non-minimal one (Myhill–Nerode) [10, Theorem 2.5.4]. Note that the minimal automaton is *trim*, that is, each state is accessible (in other words, the terminal state on a directed path from the start state).

Where M is a monoid with finite generating set Σ , we call a subset K of M *rational* if it is the image of a regular subset Q of Σ^* under the canonical monoid homomorphism $\mu : \Sigma^* \rightarrow M$. We note that this is one of a few equivalent definitions of a rational subset of a monoid, and that a subset of Σ^* is regular precisely when it is rational.

Since we want to use automata to define languages that are subsets of groups, the alphabets Σ of our automata will often be sets of the form $S^\pm = S \sqcup S^{-1}$, where S is a finite generating set of a related group.

It is well known that if a subset L of $(S^\pm)^*$ is regular, its *free reduction* $\bar{L} := \{\bar{w} : w \in L\}$ is also regular (Benois’ theorem, see [2, 1]).

2.3. Rational structures and automatic groups: A pair (G, L) is called a *rational structure* for a group G if $G = \langle S \rangle$ for a finite alphabet S and $L \subset (S^\pm)^*$ is a regular set where the restriction $\mu|_L : L \rightarrow G$ is a surjection. We note that if (G, L) is a rational structure for G , then $(G, \bar{L})$ is also a rational structure, with $\bar{L} \subset F(S)$.

Given a rational structure (G, L) and a subset K of G , we define the set of *L -representatives* of K to be the subset $L \cap \mu^{-1}(K)$ of $(S^\pm)^*$ and the set of *reduced L -representatives* of K to be the subset $\bar{L} \cap \pi^{-1}(K)$ of $F(S)$; we observe that the images the two sets in G must both contain K , since L maps onto G . We say that the subset K is *L -recognisable*¹ if its L -representatives form a regular subset of $(S^\pm)^*$ (equivalently, its reduced L -representatives form a regular subset of $F(S)$); we abbreviate $(S^\pm)^*$ -recognisable as recognisable. Fig. 1 illustrates the set of L -representatives of a rational subset $K = \mu(Q)$ of a group, where Q is a regular subset of $(S^\pm)^*$.

Following [6], we call a group G *automatic* if a rational structure (G, L) exists, which is additionally equipped with a collection of finite state automata $\{\mathcal{M}_s\}$ for $s \in S^\pm \cup \{1_G\}$ called *multipliers*, that accept strings corresponding to pairs of words (w_1, w_2) in $L \times L$ such that $w_1 s =_G w_2$ in the group.

Let G be a group with rational structure (G, L) . A subset K of G has *decidable membership problem* if there exists an algorithm for deciding whether an element $g \in G$ lies in K . The following proposition derives from [12, Proposition 3.13].

Proposition 2.1. *Let $G = \langle S \rangle$ be a group with automatic structure (G, L) , and let K be a rational subset of G . If K is L -recognisable, then K has decidable membership problem.*

Proof. Let $h \in G$. Lemma 3.10 of [12] implies that it is possible algorithmically to construct an automaton B_h which accepts precisely the L -representatives of h . If K is L -recognisable, then there exists an automaton A which accepts the L -representatives of K . Now $h \in K$ if and only if the languages of A and B_h have non-empty intersection. $\square$

¹ L -recognisable subsets first appear in [8], where they are called L -rational.

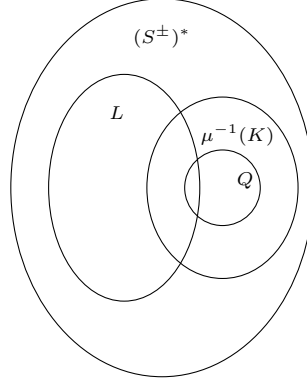


FIG. 1. L -representatives of a rational subset: L and Q are regular subsets of $(S^\pm)^*$ and the projection $\mu: L \rightarrow G$ makes (G, L) a rational structure. The subset $K = \mu(Q)$ is rational in G , and its set of L -representatives is $\mu^{-1}(K) \cap L$ which is not, in general, either regular or equal to $Q \cap L$.

We note that while Proposition 2.1 establishes that an algorithm exists to decide membership of K , it does not enable us to run that algorithm unless we can first calculate the automaton A whose language is the set of L -representatives of K . With this in mind we say that a decision problem is *constructively decidable* if it is decidable and we have an effective means to construct an algorithm to solve it.

2.4. Conventions for Cayley graphs and paths: Given a group $G = \langle S \rangle$, the *Cayley graph* of G with respect to S will be denoted by $\Gamma(G, S)$ (or just $\Gamma(G)$ if the generating set S is understood). The graph is directed and labelled. Its vertex set $V\Gamma$ is in correspondence with G and the label $l(v)$ of a vertex v is the corresponding element of G . For each $v \in V\Gamma$, and each $s \in S^\pm$, there is a directed edge $e \in E\Gamma \subseteq V\Gamma \times V\Gamma$ with label s , joining v to the vertex with label $l(v)s$.

By a path (of length $n \geq 0$) in the Cayley graph, we mean a directed graph homomorphism $\gamma: P_n \rightarrow \Gamma(G, S)$ from the path graph P_n into the Cayley graph. We often think of γ as its image in $\Gamma(G, S)$, rather than as a homomorphism.

For each edge $e = (u, v)$ in the Cayley graph, labelled by a generator s , the graph has an inverse edge $e^{-1} = (v, u)$ with label s^{-1} ; similarly, given a path γ in the graph from a vertex u to a vertex v we can assign to it a label $l(\gamma)$ in $(S^\pm)^*$ given by the concatenation of the labels of edges in the path. Each such γ has a natural inverse path γ^{-1} whose label $l(\gamma^{-1})$ is the formal inverse $l(\gamma)^{-1}$ of $l(\gamma)$.

Since $V\Gamma$ is in bijection with G through the labelling function, we will often abuse notation and simply refer to a vertex v by its label $g = l(v)$. If the initial vertex for a path is fixed (e.g. the path starts at 1_G) the set of words in $(S^\pm)^*$ is in bijection with the set of (not necessarily reduced) paths in the Cayley graph, so we will often interchangeably talk about a path γ or its label $w = l(\gamma)$.

Note that the length n of the path $\gamma: P_n \rightarrow \Gamma(G, S)$ with label $l(\gamma) = w$ coincides with length $|w|$ of the word w in the free monoid. Given an index $0 \leq i \leq |w|$, we

denote by w_i the i -th vertex of the path and by $w(i)$ the prefix of w containing its first i letters, that is, $w(i) = \gamma|_{P_i}$. When γ is a path starting in 1_G , this means that $w_i = \mu(w(i))$.

Lastly, taking the usual graph metric where each edge has length 1 we can think of the Cayley graph of a group G , and therefore of G itself, as a metric space whose metric will be denoted by $d: G \times G \rightarrow \mathbb{N}$ throughout this text.

2.5. Convexity notions: Taking into account the bijection between a group $G = \langle S \rangle$ and the vertices of the graph $\Gamma(G, S)$, it is natural to define a subset $K \subset G$ to be *convex* when, for any pair of elements $g, h \in K$, the vertices of any geodesic in $\Gamma(G, S)$ between g and h are all contained in K .

In practice, it is useful to work with a weaker condition than convexity: we define a subset K of G to be *quasi-convex* with constant k if each geodesic in $\Gamma(G, S)$ connecting elements g and h of K is contained in a k -neighbourhood of K [4, III.Γ.3]; we say that the subset is quasi-convex if such a constant k exists. In particular, the study of quasi-convex subgroups has been fruitful in the last decades (see [4, 5, 3]).

In the case where the group G has a rational structure $L \subset (S^\pm)^*$, a more general concept is defined to work with subgroups of G : a subset K is said to be *L -quasi-convex* with constant k if, for any $g \in K$, every path in $\Gamma(G, S)$ from 1_G to g defined by a word in L (in other words, every L -representative of K) lies in the k -neighbourhood of K [8, 12]. Note that this definition is particularly powerful when K is a subgroup, as closure under inversion allows us to translate any path between elements g and h to a path between 1_G and the element $g^{-1}h$ in the subgroup. This is not the case for other subsets, e.g. rational subsets. In the case of subgroups, L -quasi-convexity is equivalent to L -recognisability [8, Theorem 2.2].

Lastly, to point out some relations between quasi-convexity and L -quasi-convexity, let L_{geod} be the language of geodesics in $\Gamma(G, S)$. If $L = L_{\text{geod}}$, then a subgroup is quasi-convex if and only if it is L -quasi-convex; if $L_{\text{geod}} \subseteq L$, then an L -quasi-convex subgroup must be quasi-convex; if $L \subseteq L_{\text{geod}}$ and K is a quasi-convex subset containing 1_G , then K must be L -quasi-convex.

As we noted, the definition of L -quasi-convexity is very useful when working with subgroups but a generalisation is needed to work with the rational subsets considered in this article: this motivates our definition of L -proximity in Section 3.

3. SETUP

Let (G, L) be a rational structure for a finitely presented group $G = \langle S | R \rangle$ and let $\mu: (S^\pm)^* \rightarrow G$ be the projection from the free monoid to G . Given $Q \subseteq (S^\pm)^*$ with $K = \mu(Q)$ a rational subset of G , our goal is to build an automaton that accepts the set of L -representatives of K .

This will not always be possible (note that subgroups are rational subsets, and the subgroup membership problem is not always decidable), but if K is L -proximate (as defined in Definition 3.1) then there exists a procedure that eventually produces such an automaton, as we will see in Section 4. If the rational subset is a submonoid given by a finite generating set, then a weaker hypothesis of weak proximity will be sufficient (as defined in Definition 3.2).

The definition of L -proximity involves not only a subset K of G but a regular subset of $(S^\pm)^*$ mapped by μ onto K .

Definition 3.1. Let (G, L) be a rational structure for a group $G = \langle S \rangle$, and let Q be a regular subset of $(S^\pm)^*$. We say that Q is *L-proximate* over G with constants k, c if for every word w in $(S^\pm)^*$ that is an L -representative of an element in $\mu(Q)$, there exists a word $w' \in Q$ with $\mu(w) = \mu(w')$ such that the paths in the Cayley graph of G based at 1_G and labelled w and w' , *k-fellow travel up to at most c-reparametrisation*. That is to say, there exists a function $f: [0, |w|] \rightarrow [0, |w'|]$ satisfying $f(0) = 0$, $f(|w|) = |w'|$ and $0 \leq f(i+1) - f(i) \leq c$, and such that, using the distance d in the Cayley graph $\Gamma(G, S)$, $d(w_i, w'_{f(i)}) \leq k$ for all $0 \leq i \leq |w'|$. (See Figure Fig. 2.)

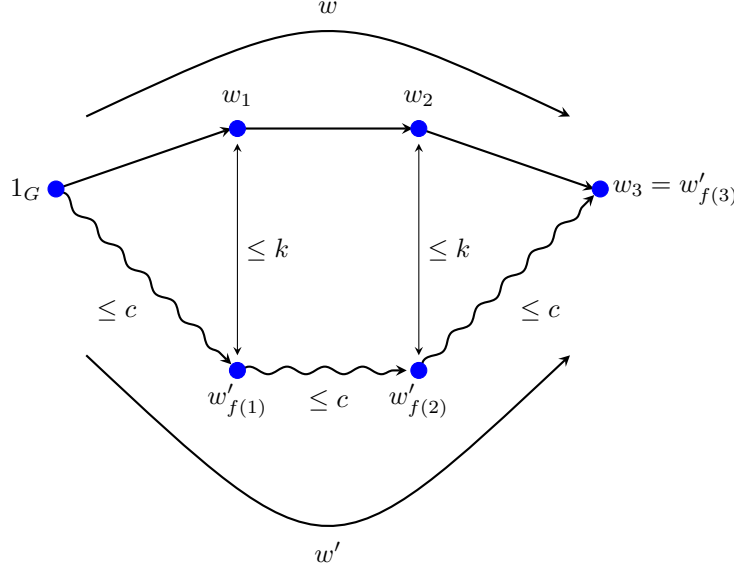


FIG. 2. Illustration of the paths in the Cayley graph defined in Definition 3.1.

In particular, w and w' *asynchronously fellow travel* (see [6] for a definition), with a (linear) restriction on their asynchronicity. A weaker notion of convexity, useful when working with submonoids of groups, is obtained by dropping the constant c :

Definition 3.2. Let (G, L) be a rational structure for a group $G = \langle S \rangle$, and let Q be a regular subset of $(S^\pm)^*$. We say that Q is *weakly-L-proximate* over G with constant k if for every word w in $L \cap \mu^{-1}(\mu(Q))$ there is a word $w' \in Q$ and a non-decreasing map $f: [0, |w|] \rightarrow [0, |w'|]$ satisfying $\mu(w) = \mu(w')$, $f(0) = 0$, $f(|w|) = |w'|$ and, using the distance d in the Cayley graph $\Gamma(G, S)$, $d(w_i, w'_{f(i)}) \leq k$ for all $0 \leq i \leq |w'|$. In other words, the paths in the Cayley graph of G represented by w' and w must *k-fellow travel asynchronously*.

Remark 3.3. Note that we may have distinct subsets of $(S^\pm)^*$ that map onto the same subset of G , one of which is *L-proximate* (or *weakly L-proximate*) while the other is not.

Take the group $\mathbb{Z}^2 = \langle a, b \mid [a, b] \rangle$ with rational structure $L = \{a, a^{-1}, b, b^{-1}\}^*$. The sets $Q_1 := L$ and $Q_2 := \{a^i b^j : i, j \in \mathbb{Z}\}$ both map onto G , but Q_1 is L -proximate while Q_2 is not. However, as we shall see in Corollary 3.12, two finitely generated submonoids of $(S^\pm)^*$ that map to the same submonoid of G are either both (weakly) L -proximate or both not (weakly) L -proximate.

Recognising dependence on the choice of Q , we define L -proximacy for a subset K of G as follows:

Definition 3.4. Given a rational structure (G, L) for a group $G = \langle S \rangle$ and a regular subset Q of $(S^\pm)^*$ which is (weakly-) L -proximate over G with constants k, c , we say that $K = \mu(Q)$ is (weakly-) L -proximate with respect to Q , with constants k, c .

We dedicate the rest of this section to understanding the relationship between weakly- L -proximate, L -proximate and L -quasi-convex subsets. It is clear that L -proximate subsets are always weakly- L -proximate, and we shall prove below that a regular language Q that is weakly- L -proximate is also L -quasi-convex. Moreover, for a submonoid given by a finite generating set T , if T^* is weakly- L -proximate it is also L -proximate and, for finitely generated subgroups, the three notions of convexity are equivalent. Briefly, we observe the following equivalences.

- For a regular subset Q of $(S^\pm)^*$ and rational subset $\mu(Q)$ of G ,

$$\begin{aligned} Q \text{ is } L\text{-proximate} &\implies Q \text{ is weakly-}L\text{-proximate} \\ &\implies \mu(Q) \text{ is } L\text{-quasi-convex.} \end{aligned}$$
- For a finite subset T of $(S^\pm)^*$ and submonoid $\mu(T^*)$ of G ,

$$\begin{aligned} T^* \text{ is } L\text{-proximate} &\iff T^* \text{ is weakly-}L\text{-proximate} \\ &\implies \mu(T^*) \text{ is } L\text{-quasi-convex.} \end{aligned}$$
- For a finite subset T of $(S^\pm)^*$ and subgroup $\mu((T^\pm)^*)$ of G ,

$$\begin{aligned} (T^\pm)^* \text{ is } L\text{-proximate} &\iff (T^\pm)^* \text{ is weakly-}L\text{-proximate} \\ &\iff \mu((T^\pm)^*) \text{ is } L\text{-quasi-convex.} \end{aligned}$$

Proposition 3.5, Proposition 3.6 and Proposition 3.7, which follow, provide detail and justification for these observations.

Proposition 3.5. *Let $G = \langle S \rangle$ be a finitely generated group with rational structure (G, L) and let $Q \subset (S^\pm)^*$ be a regular set. If Q is weakly- L -proximate over G with constant k then it is also L -quasi-convex with constant $k' = k + |V|$, where $|V|$ denotes the number of vertices in the minimal automaton accepting Q .*

Proof. Let w be a word in $L \cap \mu^{-1}(K)$ and let w' be the associated word in Q with which we are provided because of K being weakly- L -proximate. We know that, for $0 < i < |w|$, each vertex w_i is at distance at most k from the vertex $w'_{f(i)}$ in $\Gamma(G, S)$, and if the prefix words $w'(f(i))$ were in Q this would already show that K is L -quasi-convex. Even though this is not necessarily true, something very close to this does hold, as follows. We claim that there exist words $u(i)$ for $0 < i < |w|$, each of length less than $|V|$, so that the concatenation $x(i) = w'(f(i))u(i)$ belongs to Q and therefore, if we see $x(i)$ as a path in $\Gamma(G, S)$ starting at 1_G , w_i is at distance at most $k + |V|$ from the final vertex of $x(i)$.

To see that such a word $u(i)$ exists, let $\mathcal{A}_Q$ be the minimal, trim automaton associated to Q ; note that each prefix word $w'(f(i))$ labels a path in $\mathcal{A}_Q$ starting at the initial state and finishing at a certain state v that is not necessarily accepting. Since $\mathcal{A}_Q$ is trim, there is a sequence of transitions in the automaton, starting at v and ending in an accepting state, and since $\mathcal{A}_Q$ has a finite number $|V|$ of states, this sequence of transitions can be chosen to be of length at most $|V|$. We just need to take $u(i)$ to be the word labelled by this sequence of transitions. $\square$

Proposition 3.6. *Let $G = \langle S \rangle$ be a finitely generated group and (G, L) a rational structure as above. Let T be a finite subset of $(S^\pm)^*$ such that T^* is weakly L -proximate with constant k , and let $M = \mu(T^*)$ be the induced submonoid of G . Define constants ℓ and c by*

$$\begin{aligned}\ell &= \max\{|m|_S : m \in T\}, \\ c &= \max\{|m|_T : m \in M \text{ and } |m|_S \leq 2(k + \ell) + 1\}.\end{aligned}$$

Then the submonoid M is L -proximate, with respect to $Q = T^$ (Definition 3.4), with constants $k' = k + \ell$, $c' = c\ell$.*

Proof. (Illustration in Fig. 3.) Let w be an L -representative of an element of M . Since T^* is weakly L -proximate, there exists w' in T^* such that w asynchronously k -fellow travels with w' , as described in Definition 3.2. Let f be the non-decreasing function specified in that definition. Since w' is a concatenation $t_1 \cdots t_u$ of words from T , we may consider the prefixes $t_1 \cdots t_v$ for $v \leq u$; call these the T -prefixes of w' . For each i in $[0, |w|]$, define $g(i)$ to be the least number no less than $f(i)$, such that $w'_{g(i)}$ is a T -prefix of w' . Since $d_S(w_i, w'_{f(i)}) \leq k$ and $d_S(w'_{f(i)}, w'_{g(i)}) \leq \ell$, we see that $d_S(w_i, w'_{g(i)})$ is bounded above by $k + \ell$. Now if $i < |w|$, then $w'_{g(i)}$ is a left divisor in T^* of $w'_{g(i+1)}$ (since g is a non-decreasing function). So there exists $x_{i+1} \in T^*$ such that $w'_{g(i)}x_{i+1} = w'_{g(i+1)}$, and since $d_S(w'_{g(i)}, w'_{g(i+1)}) \leq 2(k + \ell) + 1$, we may assume that x_{i+1} has T -length at most c . Let w'' be the word $x_1 \cdots x_{|w|}$ in T^* , and let w''_i be the prefix $x_1 \cdots x_i$. For each i we have $w'_{g(i)} = w''_i$, and it follows that w asynchronously $(k + \ell)$ -fellow travels with w'' . But since the word x_{i+1} has T -length at most c , we see that its S -length is at most $c\ell$, from which the result follows. $\square$

Proposition 3.7. *Let $G = \langle S \rangle$ be a finitely generated group and (G, L) a rational structure. Let T be a finite subset of $(S^\pm)^*$, and let H be the subgroup of G generated by T . Define constants ℓ and c by*

$$\begin{aligned}\ell &= \max\{|h|_S : h \in T\}, \\ c &= \max\{|h|_T : h \in H \text{ and } |h|_S \leq 2k + 1\}.\end{aligned}$$

If H is L -quasi-convex with constant k , then H is L -proximate with respect to $Q = (T^\pm)^$, with constants k and $c' = c\ell$.*

Proof. Let w be an L -representative of an element of H . For $j \leq |w|$ (following the conventions of Section 2.4) let $w(j)$ be the prefix of w of length j . Then $w_j = \mu(w(j))$ lies in the k -neighbourhood of H in $\Gamma(G, S)$. Let h_j be an element of H such that $d_S(w_j, h_j) \leq k$, with $h_0 = 1$ and $h_{|w|} = \mu(w)$. For $1 \leq j \leq |w|$, define t_j to be a word of minimal T -length in $(T^\pm)^*$ such that $\mu(t_j) = h_{j-1}^{-1}h_j$. Then setting $q_j = t_1 \cdots t_j$ for each j (with $q_0 = \epsilon$), we see that $\mu(q_j) = h_j$ for all

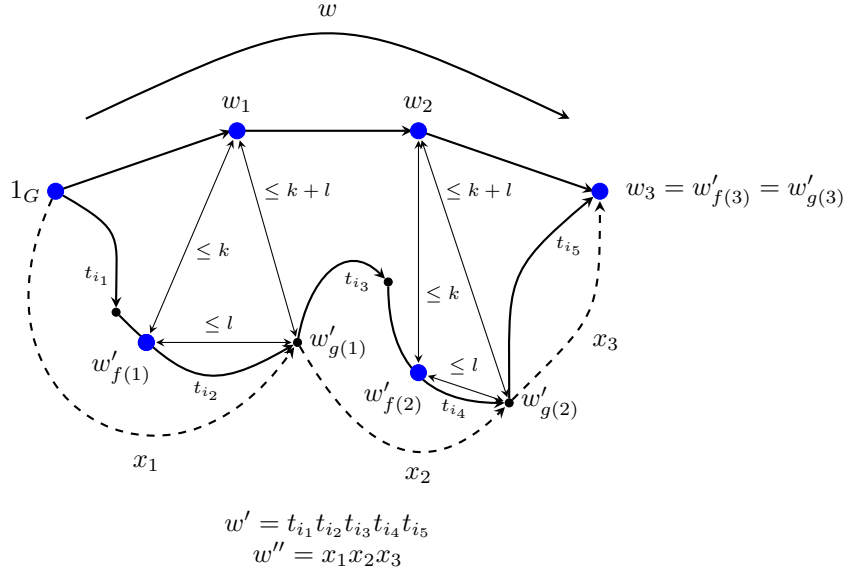


FIG. 3. The paths defined in the proof of Proposition 3.6.

j . Since $d_S(h_j, h_{j+1}) \leq 2k + 1$, we see that $|h_j^{-1}h_{j+1}|_S \leq 2k + 1$, and hence that $|h_j^{-1}h_{j+1}|_T \leq c$. Since the elements t_j have minimal T -length by assumption, it follows that $|t_j|_S \leq c\ell$ for all j . So we see that w k -fellow travels with $q_{|w|}$ up to at most c' -reparametrization, and so H is L -proximate with constants k and c' . $\square$

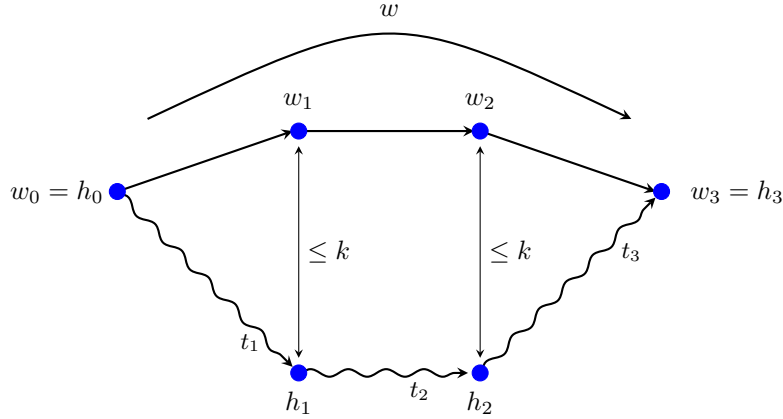


FIG. 4. The paths defined in Proposition 3.7

At the end of Section 2 we pointed out that a subgroup is L -quasi-convex if and only if it is L -recognisable. Similarly, we have the following.

Proposition 3.8. *Given a group $G = \langle S \rangle$ and a rational structure (G, L) , where $L \subseteq (S^\pm)^*$, a rational subset K of G is L -proximate, with respect to some regular subset of $(S^\pm)^*$, if and only if it is L -recognisable.*

One direction is immediate, as L -recognisability means that $L \cap \mu^{-1}(K)$ is a regular language Q , and Q is by definition L -proximate with constants $k = 0$, $c = 1$. The other direction of the statement will be proved constructively in Proposition 4.10. As a consequence of this result, Proposition 3.5 and Proposition 3.6 we have the following.

Proposition 3.9. *Let $G = \langle S \rangle$ be a group with rational structure (G, L) , where $L \subseteq (S^\pm)^*$ and let T be a finite subset of $(S^\pm)^*$. Then the submonoid $\mu(T^*)$ of G is weakly- L -proximate, with respect to some regular subset of $(S^\pm)^*$, if and only if it is L -recognisable.*

Question 3.10. Is it possible to replace “with respect to some regular subset of $(S^\pm)^*$ ” by “with respect to T^* ” in the Lemma above? It follows from Proposition 3.8 that this works for the only if part of the statement but we do not know whether the property that $\mu(T^*)$ is L -recognisable and weakly- L -proximate, with respect to some regular set Q , is sufficient to ensure that it is weakly- L -proximate with respect to T^* .

Before we move on to the next section, we give an equivalent definition for weakly- L -proximate submonoids, which clarifies the similarity to classical L -quasi-convexity and the independence on generating set.

Proposition 3.11. *Given a finitely generated group $G = \langle S \rangle$ with rational structure (G, L) and a finite set $T \subset (S^\pm)^*$, the language T^* (and therefore the submonoid $M = \mu(T^*)$ of G) is weakly- L -proximate if and only if there is a constant k such that for every $w \in L \cap \mu^{-1}(M)$ there are elements $m_1, \dots, m_{|w|}$ in M such that, in $\Gamma(G, S)$, m_i is at distance at most k from the vertex w_i , $m_{|w|} = w$ and $m_{i+1} \in m_i M$.*

Proof. Recall that $w = w_{|w|}$. If such elements $m_1, \dots, m_{|w|}$ exist, let n_1 be a T^* representative of m_1 , and let n_{j+1} be a word in T^* such that $m_{j+1} =_M m_j n_{j+1}$, for $1 \leq j \leq |w| - 1$; which exist as $m_{j+1} \in m_j M$. Then the word $n_1 n_2 \cdots n_i$ is in T^* and is mapped by μ to m_i , for $1 \leq i \leq |w|$, whence $w =_M n_1 n_2 \cdots n_{|w|}$ satisfies the weak- L -proximate condition, with constant k .

Conversely, if G is weakly- L -proximate, with constant k , let $l := \max\{|t|_S : t \in T\}$ and let $k' = k + l$. If w' is the word corresponding to w , from Definition 3.2, then using the argument given in the sketch of the proof of Proposition 3.6, it follows that elements $m_1, \dots, m_{|w|}$ of M exist, such that the distance between w_i and m_i is at most k' . $\square$

The following is an immediate consequence of Proposition 3.11.

Corollary 3.12. *In a group with rational structure (G, L) , if the images in G of finite subsets T and U of $(S^\pm)^*$ generate the same submonoid of G , then T^* is (weakly) L -proximate precisely when U^* is.*

4. A FOLDING PROCEDURE FOR RATIONAL SUBSETS

Throughout this section, we adopt the convention that a finitely presented group is given by a finite presentation $G = \langle S \mid R \rangle$ for which the set R of relators is closed under taking inverses and cyclic permutation of words.

4.1. Rewriting. In the next paragraphs we define a finite rewriting system, parallel and similar to the one presented in [12, §4], whose aim is to reflect the behaviour of the algorithm that will be introduced later in the section in Construction 4.7.

Let $G = \langle S \mid R \rangle$ be a finitely presented group and $\tilde{D}_R, \tilde{D}_{\text{red}}$ be the following rewriting systems on $(S^\pm)^*$, where $R_0 = \{ss^{-1} \mid s \in S^\pm\}$:

$$\begin{aligned}\tilde{D}_R &= \{\varepsilon \rightarrow t \mid t \in R \cup R_0\}, \\ \tilde{D}_{\text{red}} &= \{r \rightarrow \varepsilon \mid r \in R_0\}.\end{aligned}$$

Given words w, w' in $(S^\pm)^*$, we say that w' is one $\tilde{D}_R$ -rewriting step away from w if there exists an integer $n \geq 1$ and words $u^{(i)}$ such that w can be written as the concatenation

$$w = u^{(1)}u^{(2)} \dots u^{(n)},$$

(where some of the $u^{(i)}$ may be the empty word) and

$$w' = u^{(1)}t^{(1)}u^{(2)} \dots t^{(n-1)}u^{(n)}$$

so that, for each i , $\varepsilon \rightarrow t^{(i)}$ is a rule in $\tilde{D}_R$. We denote this by $w \xrightarrow{R} w'$. Note that, by allowing $n = 1$ in the definition of a $\tilde{D}_R$ -rewriting step, we have that $w \xrightarrow{R} w$, and by allowing $u^{(i)} = \varepsilon$ we allow insertion of multiple relations in the same position in w and at its beginning or end. If there are words $w_{(1)}, \dots, w_{(k-1)}$ such that

$$w \xrightarrow{R} w_{(1)} \xrightarrow{R} \dots \xrightarrow{R} w_{(k-1)} \xrightarrow{R} w',$$

then we say that w is k $\tilde{D}_R$ -rewriting steps away from w' and we write $w \xrightarrow[k]{R} w'$. We write $w \xrightarrow{*}{R} w'$ to describe that w' is obtained from w in 0 or more $\tilde{D}_R$ -rewriting steps. Using the rewriting system $\tilde{D}_{\text{red}}$, we similarly define $\tilde{D}_{\text{red}}$ -rewriting steps together with the notation

$$w \xrightarrow[\text{red}]{} w', w \xrightarrow[\text{red}]{k} w' \text{ and } w \xrightarrow[\text{red}]{*} w'.$$

Lemma 4.1. *If w, u and w' are elements of $(S^\pm)^*$ such that $w \xrightarrow[\text{red}]{} u \xrightarrow{R} w'$, then there exists a word $v \in (S^\pm)^*$ such that $w \xrightarrow{R} v \xrightarrow[\text{red}]{} w'$.*

Rather than proving this in general we exhibit an example. The (rather tedious) general proof follows the same lines. For example, suppose $w = ass^{-1}btt^{-1}c$, where $s, t \in S$ and $a, b, c \in (S^\pm)^*$, with $u = abc$ and then that $u = defg$, where $d, e, f, g \in (S^\pm)^*$, and $w' = d\alpha e\beta f\gamma g$, where $\alpha, \beta, \gamma \in R \cup \{ss^{-1} \mid s \in S\}$. Then we may refine the factors of u to allow either factorisation. If, for instance, $a = a_1a_2$ and $b = b_1b_2$ where $d = a_1$, $e = a_2b_1$, $f = b_2$ and $g = c$ then

$$\begin{aligned}u &= (a_1a_2)(b_1b_2)c = a_1(a_2b_1)b_2c, \\ w &= (a_1a_2)ss^{-1}(b_1b_2)tt^{-1}c = a_1(a_2ss^{-1}b_1)(b_2tt^{-1})c\end{aligned}$$

and

$$w' = a_1\alpha(a_2b_1)\beta b_2\gamma c.$$

Defining $v = a_1\alpha a_2ss^{-1}b_1\beta b_2tt^{-1}\gamma c$ we then have $w \xrightarrow{R} v \xrightarrow[\text{red}]{} w'$, as required.

As a consequence of Lemma 4.1:

Lemma 4.2. *If w and w' are words representing the same element in G , then w' can be obtained from w by applying a sequence of $\tilde{D}_R$ rewrites followed by $\tilde{D}_{\text{red}}$ rewrites.*

Definition 4.3. Given a finitely presented group $G = \langle S | R \rangle$, and words w, w' in $(S^\pm)^*$ we say that w' is one step away from w if there exists an intermediate word w'' such that

$$w \xrightarrow[R]{\quad} w'' \xrightarrow[\text{red}]{\quad} w'.$$

We say that w' is k steps away from w if there are words $w_{(0)} = w, w_{(1)}, \dots, w_{(k)} = w'$ such that $w_{(i)}$ is one step away from $w_{(i-1)}$ for $i = 1, \dots, k$.

Lemma 4.4. *Let w and w' be words in $(S^\pm)^*$. Then w' is k steps away from w if and only if there exists an intermediate word w'' such that*

$$w \xrightarrow[R]{\quad} w'' \xrightarrow[\text{red}]{\quad} w'.$$

Proof. If w' is k steps away from w then, by repeatedly applying the result of Lemma 4.1 a rewriting of the required form may be obtained. The converse follows directly from the definition. $\square$

Corollary 4.5. *If w and w' are words representing the same element in G , then there exists k such that w' is k steps away from w .*

Proof. This follows immediately from Lemma 4.2 and Lemma 4.4. $\square$

4.2. The procedure. Now, we describe the procedure that will allow us to recognise all of the L -representatives of a rational subset K of a group $G = \langle S \rangle$, starting from an automaton that recognises a rational subset Q of $(S^\pm)^*$ with $\mu(Q) = K$.

A key step in our procedure *folds* a finite state automaton $\mathcal{A} = (S^\pm, V, E, A, v_0)$ to produce a finite state automaton $\mathcal{A}'$ whose language is the *folding* of $L(\mathcal{A})$.

Definition 4.6. The *folding* of a subset L of $(S^\pm)^*$ is the smallest subset L' of $(S^\pm)^*$ containing L for which, if $uxx^{-1}v \in L'$ then $uv \in L'$.

By Benois' theorem [2, 1], the folding L' of a regular language $L = L(\mathcal{A})$ is regular, and an automaton $\mathcal{A}'$ to recognise it can be constructed using an algorithm described in [1]. As it forms a central part of our procedure we briefly outline the operation of this algorithm. Given a finite state automaton $M = (S^\pm \cup \{\varepsilon\}, V, E, A, v_0)$, which may have ε transitions, the algorithm constructs a new automaton M' , as follows. Regard M as a directed, labelled graph. If p and q are vertices of M and, for some $a \in S^\pm$, there is a path in M from p to q with label aa^{-1} , and no path from p to q with label equal to 1 in the free monoid $(S^\pm)^*$, then an edge from p to q , labelled ε , is added to M . This is repeated over all pairs $(p, q) \in V \times V$ to give an automaton M_1 . The entire process is then repeated, starting with M_1 in place of M , to construct M_2 . As no new vertices are added and the only new edges added have label ε , eventually the process stabilises with $M_n = M_{n+1} = M'$; at which point $L(M') = L(M)'$ and we call M' the *folding* of M . For details see [1]. We call this algorithm *Benois' algorithm*.

Construction 4.7. Let $G = \langle S | R \rangle$ be a finitely presented group, with R closed under taking inverses, and let $\mathcal{A}_0$ be an automaton over the alphabet $S^\pm$ recognising a regular language $Q \subseteq (S^\pm)^*$. For each $n > 0$, we construct $\mathcal{A}_n$ from $\mathcal{A}_{n-1}$ by performing the following sequence of two steps:

- (1) At each vertex of $\mathcal{A}_{n-1}$ add a cycle corresponding to each relator of R as well as a cycle of length two corresponding to each ss^{-1} for $s \in S^\pm$, and hence obtain an automaton $\mathcal{A}_n^0$.
- (2) Use Benois' algorithm to construct an automaton $\mathcal{A}_n^1$ by folding $\mathcal{A}_n^0$.
- (3) Determinise $\mathcal{A}_n^1$ to obtain $\mathcal{A}_n$.

Proposition 4.8. *A word $w \in (S^\pm)^*$ is recognised by the automaton $\mathcal{A}_n$ if and only if w is at most n steps away from a word accepted by $\mathcal{A}_0$.*

Proof. By definition, $\mathcal{A}_n$ is the automaton obtained by folding and determinising $\mathcal{A}_n^0$, which means that the language accepted by $\mathcal{A}_n$ is the set of all words that can be obtained through free reductions from words accepted by $\mathcal{A}_n^0$. Consequently, a word w' is accepted by $\mathcal{A}_n$ if and only if there is a word w'' accepted by $\mathcal{A}_n^0$ such that $w'' \xrightarrow[\text{red}]{*} w'$.

Take now an arbitrary word w'' accepted by $\mathcal{A}_n^0$. It is clear from the construction that we can write it as

$$w'' = r^{(0)} w_1 r^{(1)} w_2 \dots w_m r^{(m)},$$

where each word $r^{(i)}$ is in the language $(R \cup R_0)^*$ (where $R_0 = \{ss^{-1} \mid s \in S^\pm\}$, as above) and $w = w_1 w_2 \dots w_m$ is a word of length m accepted by $\mathcal{A}_{n-1}$. This means exactly that w'' is at most one R -rewriting step away from w , so altogether we deduce that

$$w \xrightarrow[R]{*} w'' \xrightarrow[\text{red}]{*} w'.$$

This means that w' in $L(\mathcal{A}_n)$ is one step away from a word w in $L(\mathcal{A}_{n-1})$ and, by induction, we have that the words accepted by $\mathcal{A}_n$ are at most n rewriting steps away from those accepted by $\mathcal{A}_0$.

The converse also follows by induction from the definition of the rewriting systems, since $\tilde{D}_R$ emulates step (1) and $\tilde{D}_{\text{red}}$ emulates step (2) in Construction 4.7: if a word w' is at most one step away from a word w accepted by $\mathcal{A}_{n-1}$ it means that we can find a word w'' (possibly equal to w) such that

$$w \xrightarrow[R]{*} w'' \xrightarrow[\text{red}]{*} w'.$$

Applying Construction 4.7, we see that the word w'' is accepted by $\mathcal{A}_n^0$ and so the word w' is accepted by $\mathcal{A}_n$. $\square$

Corollary 4.9. *Let $\langle S|R \rangle$ be a finitely presented group and $Q_0 \subseteq (S^\pm)^*$ the regular language accepted by some automaton $\mathcal{A}_0$. Denote by Q_n , $n \in \mathbb{N}$, the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above. Then, $\bigcup_{n \in \mathbb{N}} Q_n = \mu^{-1}(\mu(Q_0))$.*

Proof. From Proposition 4.8, we see that $\mu(Q_n) = \mu(Q_0)$ for all $n \in \mathbb{N}$, and it is clear that for any $w \in \mu^{-1}(\mu(Q_0))$, there exists $n \in \mathbb{N}$ such that w can be obtained in n steps from a word of Q_0 . $\square$

As stated above, for an L -proximate rational subset K , our procedure will eventually construct an automaton which recognises all L -representatives of K .

Proposition 4.10. *Let (G, L) be a rational structure for a finitely presented group $G = \langle S|R \rangle$ and let $Q_0 \subseteq (S^\pm)^*$ be the regular language accepted by an automaton $\mathcal{A}_0$. Assume that Q_0 is L -proximate with constants c, k (see Definition 3.1) and let $K = \mu(Q_0)$. Then, there exists $n \in \mathbb{N}$ such that the full set of L -representatives of K*

is contained in Q_n , the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above. In particular, K is L -recognisable.

The following lemma, which follows easily from the definitions, will aid the proof of Proposition 4.10.

Lemma 4.11. *Let $G = \langle S \mid R \rangle$ be a finitely presented group, let $m, n \in \mathbb{N}$ be strictly positive integers, and let $w_1, \dots, w_m, w'_1, \dots, w'_m \in (S^\pm)^*$ be words such that each w'_i can be obtained in at most n steps from w_i . Then, $w'_1 w'_2 \dots w'_m$ can be obtained in n steps from $w_1 w_2 \dots w_m$.*

Proof of Proposition 4.10. Corollary 4.5 tells us that whenever u and v are words in $(S^\pm)^*$ with $u =_G v$, there is some minimal $k_{u,v} \in \mathbb{N}$ such that u is $k_{u,v}$ steps from v . Define $n \in \mathbb{N}$ by

$$n = \max\{k_{u,v} \mid u =_G v, |u| \leq 2k + 1, |v| \leq c\}$$

This maximum exists since the elements u and v under consideration lie in a bounded ball in the Cayley graph of G with respect to S . We will show that the set of L -representatives $J = L \cap \mu^{-1}(K)$ is a subset of Q_n .

Let w be an arbitrary word in the language J . By assumption, there exists a word $w' \in Q_0$ with $w =_G w'$ such that the paths in the Cayley graph of G corresponding to the words w and w' k -fellow travel, up to c -reparametrisation. Let $f: [0, |w|] \rightarrow [0, |w'|]$ be the reparametrisation map so that, for the paths labelled by w and w' , following the notation defined in Section 2, we have $0 \leq f(i+1) - f(i) \leq c$ and $d(w_i, w'_{f(i)}) \leq k$ for all $0 \leq i \leq |w|$.

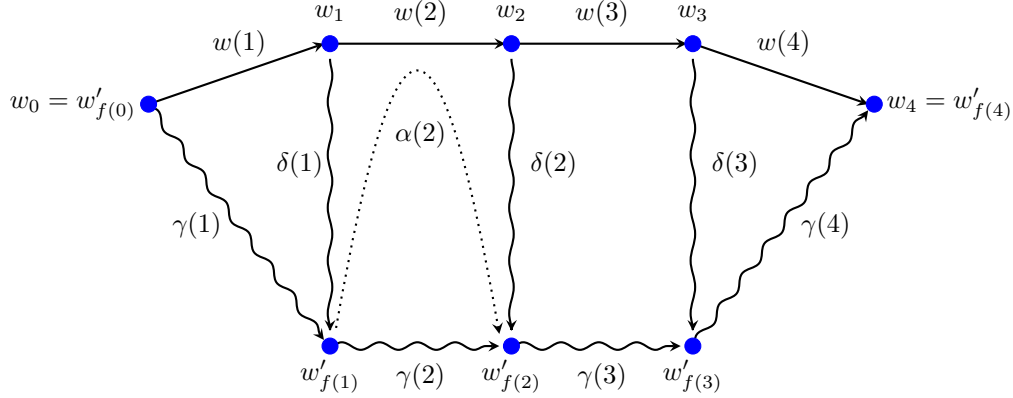


FIG. 5. Notation for the proof of Proposition 4.10

See Fig. 5 for an illustration of the following paragraph. Following Section 2 denote by $w(1), \dots, w(|w|)$ the letters in w , which we can think of as S -labelled, 1-edge subpaths of w in the Cayley graph. Let $\gamma(1), \gamma(2), \dots, \gamma(|w|)$ be the subwords of w' defined by respectively taking its first $f(1)$ letters, then the next $f(2) - f(1)$ letters, etc. until finally making $\gamma(|w|)$ out of the last $f(|w|) - f(|w| - 1)$ letters of w' . Each $\gamma(i)$ can be thought of as a subpath of w' in the Cayley graph of length less than or equal to c . Now, let $\delta(i) \in (S^\pm)^*$ be a word representing a geodesic path between the vertices w_i and $w'_{f(i)}$, for $i = 0, \dots, |w|$. Note that such words

have length less or equal than k , and that $\delta(0), \delta(|w|)$ are trivial paths. Lastly, define $\alpha(i)$ to be the path in the Cayley graph (and the word in $(S^\pm)^*$) defined by concatenating the paths $\delta^{-1}(i-1) \cdot w(i) \cdot \delta(i)$, where δ^{-1} denotes the natural inverse path. The paths $\alpha_1, \dots, \alpha_{|w|}$ all have length less than or equal to $2k+1$.

Note that each of the paths $\alpha(i)$ defines a word equivalent to $\gamma(i)$ in G , and because $|\alpha(i)| \leq 2k+1$ and $\gamma(i) \leq c$ the word $\alpha(i)$ is at most n steps away from $\gamma(i)$. By Lemma 4.11 we conclude that the word defined by $\alpha = \alpha(1) \cdot \dots \cdot \alpha(|w|)$ is at most n steps away from $w' = \gamma(1) \cdot \dots \cdot \gamma(|w|)$ and, since w' is in Q_0 , this means that Q_n contains the word defined by α . Furthermore, $\alpha = \alpha(1) \cdot \dots \cdot \alpha(|w|)$ is freely reducible to $w = w(1) \cdot \dots \cdot w(|w|)$ which means that w is also accepted by Q_n (since its language is closed under free reductions).

For the final statement; since the intersection of two regular languages is regular, it follows that $\mu^{-1}(K) \cap L = Q_n \cap L$ is regular, so K is L -recognisable. $\square$

Definition 4.12. Given a set T of words over $S^\pm$, we define the *flower automaton* $\text{Flower}(T)$, which accepts T^* , to be the automaton over $S^\pm$ formed by attaching disjoint directed loops, each labelled by an element $t \in T$ to a vertex s_0 that is the both the start state and the single accepting state.

As a consequence of Proposition 4.10 and Proposition 3.6, we have the following result.

Proposition 4.13. *Let (G, L) be a rational structure for a finitely presented group $G = \langle S | R \rangle$, let $T \subseteq (S^\pm)^*$ be a finite set of words and consider the regular language $Q_0 := T^*$ accepted by the flower automaton $\mathcal{A}_0 := \text{Flower}(T^*)$ (Definition 4.12). If the submonoid $M = \mu(Q_0)$ is weakly- L -proximate with respect to Q_0 , with constant k (see Definition 3.2) then there exists $n \in \mathbb{N}$ such that the full set of L -representatives of M is contained in Q_n , the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above.*

5. APPLICATION TO SUBMONOIDS OF AUTOMATIC GROUPS

In this section we use the argument of Kharlampovich, Myasnikov and Weil [12] to turn the procedure of the previous section into a partial algorithm which will construct an automaton accepting the set of L -representatives of a given finitely generated monoid of an automatic group. The main ingredient is a test to show whether or not the process has completed. The test works for submonoids because the automaton corresponding to a finite generating set of a monoid has start state equal to its unique accept state. As we shall see, this means that the following lemma, from [12], gives rise to the required test. Throughout the section, the conventions for symmetric presentations of a group remain as in Section 4.

For the remainder of this section let G be an automatic group with generating set S and $L \subseteq (S^\pm)^*$ be the language of this automatic structure.

Lemma 5.1 (Lemma 4.11 in [12]). *Given $h \in F(S)$ and a regular language Q contained in L , there exists an algorithm that constructs an automaton accepting the set $\mathcal{M}_h(Q) = L \cap \mu^{-1}(\mu(Qh))$ of all the L -representatives of the elements of $\mu(Qh)$.*

Proposition 5.2. *Let T be a finite subset of $S^\pm$, such that the submonoid $M = \mu(T^*)$ of G generated by T is weakly- L -proximate. Let $\mathcal{A}_0 := \text{Flower}(T)$ (Definition 4.12), and let $\mathcal{A}_0, \mathcal{A}_1, \dots$ be the sequence of automata constructed in Section 4.*

Then, for $i \geq 0$, the automaton $\mathcal{A}_i$ may be effectively constructed and there is an algorithm which, on input T , outputs an integer $n \geq 0$ such that the language $L(\mathcal{A}_n)$ contains all L -representatives of M .

Proof. First we describe the algorithm and then we verify that it works as required. The algorithm begins by finding an element $w_\varepsilon \in L$ such that $\mu(w_\varepsilon) = 1_G$. As G is automatic this can be done using the multiplier automaton for ε . After this pre-processing step, for each $i \geq 0$ until it halts, the algorithm consists of 3 steps. The input at each iteration is the automaton $\mathcal{A}_i$ and we set $Q_i := L(\mathcal{A}_i)$. In the first step the algorithm checks to see that w_ε is in Q_i . If not then the algorithm passes to the third step described below. If $w_\varepsilon \in Q_i$ the algorithm carries out its second step in which it

- constructs an automaton for the language $Q'_i := L \cap Q_i$;
- constructs an automaton accepting $\mathcal{M}_t(Q'_i)$, for each $t \in \{\varepsilon\} \cup T$;
- tests to see whether or not $\mathcal{M}_t(Q'_i)$ is contained in Q_i .

If the answer to the test is “yes”, for all $t \in \{\varepsilon\} \cup T$, the algorithm halts and outputs i and $\mathcal{A}_i$. Otherwise the algorithm continues to step 3 in which Construction 4.7 is used to construct $\mathcal{A}_{i+1}$; and then moves on to iteration $i + 1$.

From Proposition 4.13, Lemma 5.1 and the standard theory of finite state automata, it follows that this procedure may be executed, and effectively constructs $\mathcal{A}_i$ at the i th iteration. To complete the proof we show that it halts and that, when it does so, the final claim holds.

First suppose that, for some $n \geq 0$, the language $Q_n := L(\mathcal{A}_n)$ contains $L \cap \mu^{-1}(M)$. Then, in particular, $w_\varepsilon \in Q_n$ and $L \cap \mu^{-1}(M) \subseteq L \cap Q_n = Q'_n$. As $\mu(Q'_n) = \mu(Q_n) = M$ and $\mu(t) \in M$, for all $t \in \{\varepsilon\} \cup T$, we have $\mu(Q'_n t) \subseteq M$ so $L \cap \mu^{-1}(\mu(Q'_n t)) \subseteq L \cap \mu^{-1}(M) \subseteq Q_n$. Hence $\mathcal{M}_t(Q'_n) \subseteq Q_n$, for all $t \in \{\varepsilon\} \cup T$.

Conversely suppose that, for some $n \geq 0$, $w_\varepsilon \in Q_n$ and, for all $t \in \{\varepsilon\} \cup T$, we have $\mathcal{M}_t(Q'_n) \subseteq Q_n$. Suppose $w \in Q_0$, so $w = t_1 \cdots t_m$, where $t_i \in T$, $m \geq 0$. If $m = 0$ then $w = \varepsilon$. As $w_\varepsilon \in Q'_n$, we have $Q'_n \neq \emptyset$, so all L -representatives of $\mu(\varepsilon) = \mu(w_\varepsilon)$ belong to $\mathcal{M}_\varepsilon(Q'_n) \subseteq Q_n$.

Now assume that $m \geq 1$, and that if u is a product of fewer than m elements of T , all L -representatives of $\mu(u)$ belong to Q_n . Then the L -representatives of $\mu(t_1 \cdots t_{m-1})$ (of which there is at least one) belong to Q'_n , so the L -representatives of $\mu(w)$ all belong to $\mathcal{M}_{t_m}(Q'_n) \subseteq Q_n$; whence all L -representatives of elements of $\mu(Q_n) = M$ belong to Q_n .

We have shown that $L \cap \mu^{-1}(M) \subseteq Q_n$ if and only if both $w_\varepsilon \in Q_n$ and $\mathcal{M}_t(Q'_n) \subseteq Q_n$, for all $t \in \{\varepsilon\} \cup T$. Therefore the algorithm halts with the required output. $\square$

Corollary 5.3. *In the notation and terminology of Proposition 5.2 an automaton recognising $L \cap \mu^{-1}(M)$ may be effectively computed and the submonoid membership problem of M is constructively decidable.*

Proof. To obtain an automaton accepting exactly $L \cap \mu^{-1}(M)$ we use a standard algorithm for an automaton for the language $L \cap L(\mathcal{A}_n)$, where $\mathcal{A}_n$ is output from the algorithm of Proposition 5.2. For the membership problem we use the argument of Proposition 2.1 and the automaton $\mathcal{A}_n$. $\square$

Remark 5.4. In general, although L -proximate rational subsets of a group G have decidable membership problem, whether this is constructive decidability is

open. For example, the proof of Proposition 2.1 breaks down if Q_0 is an arbitrary regular subset of $(S^\pm)^*$. The difficulty is to find an analogue of the generating set T and then an analogue of the statement that given that $w_\varepsilon \in Q_n$ then the fact that $\mathcal{M}_t(Q'_n) \subseteq Q_n$, for all t , implies that $L \cap \mu^{-1}(M)$ is contained in Q_n . Both of these difficulties arise when Q_0 has distinct start and final state. In the first author's thesis further analysis of the general case of L -proximate rational subsets is carried out and some (technical) conditions are identified, under which a halting test can be devised.

6. APPLICATION TO SUBMONOIDS OF SURFACE GROUPS

The goal of this section is to give examples of weakly- L -proximate submonoids that lie inside surface groups, where L is the language of geodesics relative to a standard generating set. We use tools from small cancellation theory to ensure that the necessary convexity conditions hold.

Definition 6.1. The *surface group* Σ_g of genus $g \geq 1$ is the group with presentation

$$\Sigma_g \equiv \langle a_1, b_1 \dots a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle.$$

We shall write Y_g for the set of generators in the presentation above, and r_g for the single relator.

The group Σ_g is the fundamental group of a closed, compact, orientable surface S_g of genus g . Except where otherwise stated, we assume that $g > 1$ throughout this section. In this case Σ_g is a hyperbolic group and therefore automatic, with a rational structure L given by geodesics with respect to Y_g .

Given a set R of relators over a generating set S , the *symmetrisation* R_* is the set of cyclic permutations of words in R , together with their inverses. We write R_g for the symmetrisation of $\{r_g\}$. Recall that Dehn's rewriting system for Σ_g has the following rewriting rules:

- (1) $ss^{-1} \rightarrow \varepsilon$, for a letter s in $Y_g^\pm$.
- (2) $u \rightarrow v$, for words u, v in $(Y_g^\pm)^*$ such that $uv^{-1} \in R_g$ and $|u| > |v|$.

A theorem of Dehn states that any word in $(Y_g^\pm)^*$ which represents the identity of Σ_g can be reduced to ε by repeated applications of these rewriting rules (Dehn's algorithm).

It is useful to think of the Cayley graph $\Gamma(\Sigma_g, Y_g)$ as a tessellation of the hyperbolic plane $\mathbb{H}^2$ by $4g$ -gons with edges labelled by the relator r . The second of Dehn's rewriting rules replaces any path going around more than half of a $4g$ -gon with the shorter route using the complementary edges. See Fig. 6 for some genus 2 examples.

6.1. Results from small cancellation theory. We shall need to introduce a result, Theorem 9.4, from [14]. We recall some standard definitions from small cancellation theory, which can be found in [13], or in the small cancellation theory appendix to [9].

Definition 6.2. Let $P = \langle S \mid R \rangle$ be a group presentation. A word u over S is a *piece* with respect to P if u is a common prefix of two distinct relators in R_* .

For the standard presentation of Σ_g , the pieces are precisely the 1-letter words.

Definition 6.3. Let $P = \langle S \mid R \rangle$ be a group presentation.

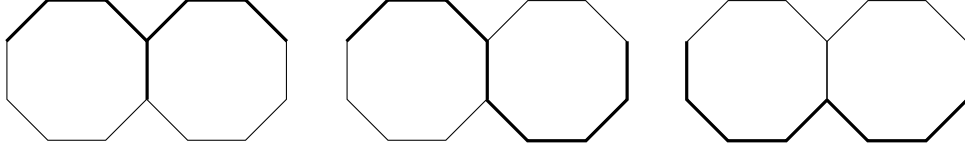


FIG. 6. Three examples of words in Σ_2 depicted as paths (in bold) along the Cayley graph of Σ_2 , which tessellates the hyperbolic plane with octagons. The leftmost path is not Dehn-reduced because it exhibits backtracking, the middle one is not Dehn-reduced because it goes around 5 edges of an octagon. The rightmost path represents a Dehn-reduced word.

- For a natural number p , we say that a P satisfies condition $C(p)$ if no element of R_* is the concatenation of fewer than p pieces.
- For $q > 2$ we say that P satisfies condition $T(q)$ if for every $\ell \in \{3, 4, \dots, q-1\}$, and for every sequence $\{r_1, r_2, \dots, r_\ell\}$ in R_* such that $r_i \neq r_{i+1}^{-1}$ for all i , and such that $r_\ell \neq r_1^{-1}$, at least one of the products $r_1 r_2, \dots, r_{\ell-1} r_\ell, r_1 r_{\ell-1}$ is freely reduced.

A presentation which satisfies both of the conditions $C(p)$ and $T(q)$ is said to be $C(p)$ – $T(q)$. We note that the standard presentation for Σ_g is $C(4g)$ – $T(4g)$; in particular, it follows that this presentation is $C(4)$ – $T(4)$.

Recall that a Van Kampen diagram for the presentation $\langle S \mid R \rangle$ of a group G is a finite, connected, contractible cell complex, embedded into $\mathbb{R}^2$, such that the edges are directed and labelled by elements of S , and such that the labels along the boundary of each 2-cell spell a element of R_* .

A *boundary cycle* of a Van Kampen diagram X is a closed path p , based at a vertex of the boundary ∂X and winding exactly once around X (with one of the two possible orientations). The label of a boundary path is a *boundary word* of X .

A pair of discs D_1 and D_2 of a Van Kampen diagram is said to be *cancelable* if the boundaries of D_1 and D_2 contain a directed edge e such that the labels of D_1 and D_2 read by starting with the edge e are equal. A Van Kampen diagram which contains no pair of cancelable discs it is said to be *reduced*. It is well known (see for example [14]) that a word $w \in (S^\pm)^*$ represents the identity element of G if and only if it is a boundary word of a Van Kampen diagram for $\langle S \mid R \rangle$.

An *arc* in the Van Kampen diagram X is a simple path in the 1-skeleton of X whose internal vertices have degree 2; an arc is *maximal* if it is not properly contained in any other arc. An arc is an *internal arc* if its interior lies in the interior of X , and a *boundary arc* if it lies entirely on the boundary of X . We notice that the labels along a maximal internal arc in a reduced Van Kampen diagram spell a maximal piece. Hence for the usual presentation of Σ_g , all maximal internal arcs have length 1. (That is to say, there are no internal vertices with degree 2.) The following definitions are taken from [14].

Definition 6.4. A *spur* is a 1-cell which is not in the boundary of a 2-cell and which is attached to the rest of the diagram at only one end. An *i-shell* is a 2-cell with exactly i maximal arcs of its boundary lying in the interior of the diagram and exactly one maximal boundary arc.

For the next definition we follow [16].

Definition 6.5. A *ladder* is a Van Kampen diagram X which is the union of a sequence of closed 1-cells and 2-cells $c_1, \dots, c_n$, such that for $1 < j < n$, there are exactly two components in $X - c_j$, such that $X - c_1$ and $X - c_n$ each has exactly one component, and such that any c_i which is a 1-cell is not contained in any other closed c_j .

A ladder can be considered as a diagram with two distinguished points x and y on its boundary, such that if P and Q are the two boundary paths from x to y , then every 2-cell has at least one edge from P and another from Q , and such that every edge not contained in a 2-cell is common to both P and Q . (We consider the paths P and Q as the rails of the ladder.)

The following result is [14, Theorem 9.4].

Proposition 6.6. *If X is a Van Kampen diagram for a $C(4)$ - $T(4)$ presentation, then one of the following holds:*

- (1) X contains at least 3 spurs or i -shells with $i \leq 2$;
- (2) X is a ladder;
- (3) X consists of a single vertex or a single 2-cell.

6.2. Dehn-reduced words and geodesics in surface groups. Note that an arbitrary word w of $(Y_g^\pm)^*$ is not necessarily reduced to a geodesic word by Dehn's algorithm. The rightmost path in Fig. 6 represents a Dehn-reduced word which is not geodesic. We shall show, using small cancellation properties of our presentation for Σ_g , that Dehn-reduced words asynchronously fellow-travel with geodesics. This gives us a tool to find weakly- L -proximate languages which describe submonoids of surface groups, and hence a class of L -recognisable submonoids.

We need the following lemma.

Lemma 6.7. *Let X be a reduced Van Kampen diagram for the surface group Σ_g . Suppose that a boundary cycle of X starting at a given vertex is labelled by a concatenation of Dehn-reduced words $w_1, \dots, w_n$. Suppose also that X has an i -shell Y with $i \leq 2g - 1$. Then the maximal arc which is the intersection of Y with the boundary of X , contains edges from two consecutive words $w_j w_{j+1}$ or $w_n w_1$.*

Proof. Since the only relator for Σ_g has length $4g$, it follows that the 2-cell Y has $4g$ edges, all of which are maximal arcs. Since at most $2g - 1$ of these edges are internal, we see that Y has at least $2g + 1$ boundary edges, and that the labels for these edges spell a subword of an element of R_g . But since each w_i is Dehn-reduced, none can contain a subword of an element of R_g of length $2g + 1$, and so the boundary edges of Y must contain labels from two consecutive words $w_i w_{i+1}$. $\square$

We can now obtain the following result.

Lemma 6.8. *Let u be a Dehn-reduced word in $(Y_g^\pm)^*$, and let v be a geodesic representing the same element of Σ_g . (So v is also Dehn-reduced.) A reduced Van Kampen diagram for the word uv^{-1} must be a single vertex, a single 2-cell, or a ladder. In particular, u asynchronously $(g + 1)$ -fellow travels with v (with upper bound of $2g$ on asynchronicity).*

Proof. We see from [16, Corollary 2.8] that if X is a diagram for a $C(4)$ - $T(4)$ presentation, and if uv^{-1} is the boundary cycle of X , then either u or v contains all of the boundary edges of a 1-shell or a 2-shell of X , or else X is a single vertex, a 2-cell, or a ladder. Let X be a Van Kampen diagram for uv^{-1} , and let u_0 and v_0

be the vertices corresponding to the start of the words u and v^{-1} respectively. By Lemma 6.7 any 1- or 2-shell of X must include edges from both u and v^{-1} , and so neither u nor v^{-1} contains all of the boundary edges for any 1- or 2-shell. So X is a vertex, a 2-cell, or a ladder.

If X is a vertex then clearly u and v fellow-travel. If X is a 2-cell then, as u and v are Dehn-reduced, both have length $2g$ and so the result holds. Assume then that X is a ladder. In this case its endpoints are u_0 and v_0 , and the rails are u and v . Each cell c_i forming the ladder must intersect both rails, and this observation, together with the fact that shells along the Dehn-reduced word u intersect the boundary of X in at most $2g$ edges, and internal arcs have length at most 1, shows that every vertex of u is at distance less than $g + 1$ from a vertex of v and that these vertices of v are separated by at most $2g - 2$ edges of v . $\square$

6.3. Constructing L -recognisable submonoids of surface groups. We start with an immediate corollary to Lemma 6.8.

Lemma 6.9. *A Dehn-reduced language Q over the generators of Σ_g is weakly- L -proximate in the surface group Σ_g . In particular, if T is a generating set for a submonoid $M \subset \Sigma_g$, and if every word in T^* is Dehn-reduced, then M is L -recognisable, and M has constructively decidable membership problem.*

Before we give examples of submonoids to which this result applies, we note an immediate way to strengthen it.

Definition 6.10. Let G be a group with presentation $\langle S | R \rangle$, and let u, v be words in $(S^\pm)^*$. We say that u yields v in a *simultaneous Dehn-reduction*, $u \xrightarrow{\text{sim}} v$, if we can write u and v as a concatenation of subwords $u = u_1 \dots u_n$ and $v = v_1 \dots v_n$ such that v_i can be obtained from u_i by applying a Dehn rewriting step. We say that u is N *simultaneous Dehn rewriting steps* from v .

Proposition 6.11. *Let Q be a language over Y_g all of whose words are at most N simultaneous Dehn rewriting steps away from being Dehn-reduced. Then Q is weakly- L -proximate with constant $k = (2N + 1)g + 1$. If $Q = T^*$, then the submonoid $M = \mu(T^*)$ is L -recognisable and has constructively decidable membership problem.*

Proof. We observe that if u yields v in a simultaneous Dehn-reduction, then u asynchronously $2g$ -fellow travels with v . Let Q' be the language of Dehn-reduced words obtainable from words in Q by at most N simultaneous Dehn-reductions. Then for every word $u \in Q$ there is $v \in Q'$ such that u asynchronously $2Ng$ -fellow travels with v . Since v is Dehn-reduced, there is a geodesic w representing the same element of G such that v asynchronously $(g + 1)$ -fellow travels with w by Lemma 6.8. So u asynchronously k -fellow travels with w , where $k = (2N + 1)g + 1$, and so Q is weakly- L -proximate with constant k .

If Q is a submonoid T^* of $(S^\pm)^*$ then Q , is L -recognisable by Proposition 3.9. $\square$

Now we see how to build a family of Dehn-reduced submonoids.

Proposition 6.12. *Let $T \subset (S^\pm)^*$ be such that*

- (1) *for all $t \in T$, t is Dehn-reduced and has length at least g ;*
- (2) *for all $t \in T$, neither its length g prefix nor its length g suffix are a prefix of any word of R_g ;*
- (3) *If a letter is an initial (final) letter of a word in T then its inverse is not a final (initial) letter of a word in T .*

Then the language T^* is Dehn-reduced. Therefore T^* is weakly- L -proximate, and the submonoid it generates is L -recognisable.

This provides a simple way of constructing Dehn-reduced submonoids of a surface group.

Example 6.13. Let $\Sigma_2 = \langle Y_2 \mid r_2 \rangle$ be the standard presentation for a genus 2 surface group, with $Y_g = \{a, b, c, d\}$ and $r_2 = aba^{-1}b^{-1}cdc^{-1}d^{-1}$. Let $T = \{w_1, w_2, w_3, w_4\}$, where

$$\begin{aligned} w_1 &= a^{-1}c u_1 cc, \\ w_2 &= bc \quad u_2 ac^{-1}, \\ w_3 &= bd^{-1} u_3 ad, \\ w_4 &= b^{-1}d u_4 c^{-1}d, \end{aligned}$$

and the words $u_1, \dots, u_4$ can be arbitrarily chosen as long as the resulting words $w_1, \dots, w_4$ remain Dehn-reduced. Then, the set T satisfies the hypothesis of Proposition 6.12 and T^* is Dehn-reduced. For a concrete example it suffices to take, among many possibilities,

$$\begin{aligned} u_1 &= d, \quad \text{so that } w_1 = a^{-1}cdc^2, \\ u_2 &= d^{-1}, \quad \text{so that } w_2 = bcd^{-1}ac^{-1}, \\ u_3 &= c^{-1}, \quad \text{so that } w_3 = bd^{-1}c^{-1}ad, \\ u_4 &= cd^{-1}, \quad \text{so that } w_4 = b^{-1}dcd^{-1}c^{-1}d. \end{aligned}$$

In recent work by Foniqi and Gray [7], the authors provide some results regarding the membership problem of certain submonoids of the surface group. More specifically, they guarantee decidability of the submonoid membership problem in a surface for Magnus submonoids [7, Theorem 4.4], for submonoids where the y -exponent of all the words in T is controlled for some $y \in Y_g$ [7, Theorem 4.1], and for certain graded submonoids [7, §6]. We note that the submonoids of our examples above are not covered by [7, Theorem 4.1] or [7, Theorem 4.4] and not obviously within the scope of results of [7, §6.3].

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